MediMatch

DATA STRUCTURES AND ALGORITHMS - 3 PROJECT

Symptom-to-Diagnosis Matcher

A clinical decision-support system that turns free-text symptoms into ranked, explainable condition matches

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Introduction

- People often describe symptoms in everyday language - "I feel feverish and my throat hurts" - not in medical terms.
- Matching such free text to possible conditions needs normalization, a knowledge base and a clear ranking method.
- MediMatch is a Java + DSA project that compares entered symptoms with a sample knowledge base and shows ranked, educational matches.
- It also suggests example follow-up assessments for the top conditions.



Educational tool
Not a substitute for a doctor

02 ABSTRACT

Abstract

The Symptom-to-Diagnosis Matcher converts free-text patient symptoms into ranked lists of probable medical conditions.

NLP normalization
Handles spelling errors, abbreviations, synonyms and varied expressions.

Knowledge base
Diseases, associated symptoms and diagnostic test information.

Weighted scoring
Transparent confidence score from relevance, frequency and importance.

Test recommendation
Minimal set of tests that best separate the top-ranked conditions.

03 PROBLEM STATEMENT

Problem Statement

Patients describe symptoms informally, while medical knowledge is structured. Bridging the two is slow, inconsistent and error-prone.

A

Unstructured input

Free text has typos, abbreviations and synonyms that simple keyword search misses.

B

Opaque results

Many tools give an answer without showing why a condition was ranked high.

C

Unnecessary tests

Without guidance, many overlapping investigations get ordered.

D

Limited access

Telemedicine, primary screening and students need quick, understandable triage support.

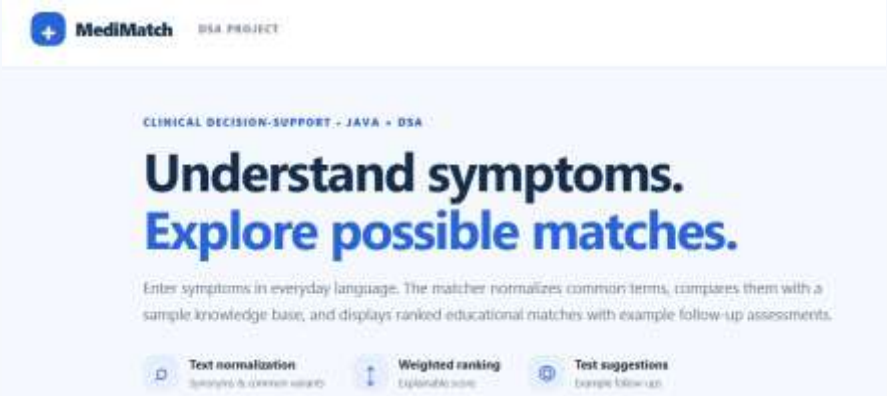
04 OBJECTIVES

Objectives

1. Accept symptoms in everyday language and normalize them to standard labels
2. Match them against a structured knowledge base of conditions
3. Rank conditions with a transparent, explainable weighted score
4. Suggest a minimal set of discriminating follow-up assessments
5. Apply core DSA concepts: HashMap, HashSet, sorting, greedy selection

05 SOLUTION

Proposed Solution



MediMatch web interface (Java back end)

Type symptoms, get ranked educational matches with the reasoning behind each score.

Text normalization
Synonyms and common variants

Weighted ranking
Explainable score per condition

Test suggestions
Example follow-up assessments

06 ALGORITHM

How the Java algorithm works

01

Normalize

Lowercase text and map phrases or synonyms to standard symptom labels.

`HashMap`

02

De-duplicate

Keep each recognized symptom once for consistent matching.

`HashSet`

03

Score & sort

Add weights for matching symptoms, compute a relative score, rank conditions.

`Weighted scoring + sorting`

04

Suggest assessments

Select a small set of follow-up tests linked to the entered symptoms.

`Greedy selection`

Input text > Normalize > De-duplicate > Score & rank > Suggest tests > Output

09 IMPACT

Applications & Benefits

1

Telemedicine

Quick preliminary triage before a consultation.

2

Primary screening

Helps staff prioritise cases at first contact.

3

Medical education

Students see how symptoms map to conditions and scores.

4

Efficiency

Suggests few, discriminating tests and avoids extras.

10 FUTURE

Limitations & Future Scope

Current limitations

- Sample knowledge base only
- Educational use, not a diagnosis
- Rule-based synonym mapping

Future scope

- Larger, clinically reviewed knowledge base
- Spelling correction and richer NLP
- Severity and duration weighting
- Multi-language input and mobile app

11 CONCLUSION

Understand symptoms. Explore possible matches.

- MediMatch combines NLP-style normalization, a knowledge base and interpretable weighted scoring.
- HashMap, HashSet, sorting and greedy selection make it fast and explainable.
- It supports preliminary triage, telemedicine, screening and medical education.

Thank you | Questions?